

17-683: Data Structures for Application Programmers

Course Syllabus

Spring 2025 Mini 4 (6 units)

Tepper Quad (TEP) 1403

Tuesdays and Thursdays 11:00am – 12:20pm

All course materials will be posted on Canvas.

Course Description

This course introduces fundamental data structures and algorithms for students with prior programming experience. Students are expected to have a basic understanding of object-oriented programming concepts, functions, loops, and arrays before enrolling. The course balances conceptual understanding with practical implementation, covering essential data structures and their associated algorithms. Students will gain hands-on experience with Java Collections (ArrayList, LinkedList, HashSet, TreeSet, HashMap, TreeMap, PriorityQueue) to reinforce theoretical concepts. The course also introduces algorithm analysis using Big O notation to evaluate efficiency and performance tradeoffs. Following a learning-by-doing approach, students will write code frequently, with multiple opportunities to implement data structures, apply algorithms to solve problems, and review concepts through practical programming exercises.

Objectives

After taking this course, students will be able to:

- Define fundamental data structures (including lists, maps, trees, stacks, queues, and heaps) and describe their theoretical properties
- Explain the logical operations and core mechanisms of common searching and sorting algorithms
- Implement standard data structures using Java with proper abstraction and encapsulation
- Analyze the performance of data structures and algorithms using the Big O notation
- Design solutions to practical problems by selecting and justifying appropriate data structures

Background and Motivation

As a Java application programmer, you may often use the Collections Framework that provides a well-designed set of interfaces and classes for storing and manipulating data. But, you may not really think about how things work under the hood.

This course is designed to take a closer look at the interfaces and classes in the Framework so that you can have a better understanding of core data structures and their major operations. As with most courses, success in this course is proportional to your engagement with the material: data structures concepts build upon each other, and regular practice with these structures is essential for succeeding in this course. Former students have reported that this course often prepares them well for technical interviews associated with software engineering positions (and adjacent roles).

Instructors

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Materials

Textbook: There is no required textbook for this course. An optional textbook is Data Structures and Algorithms in Java (2nd Edition) by Robert Lafore.

Grading & Academic Integrity Policy

You are responsible for being familiar with the university standard for academic integrity and plagiarism.

<http://www.cmu.edu/policies/student-and-student-life/academic-integrity.html>

Please see the CMU Student Handbook for information. To deter and detect plagiarism, ***we will be using the Moss system***. All homework assignments and labs are handed in using Autolab (<https://autolab.andrew.cmu.edu>).

Students caught cheating or plagiarizing will receive ***no credit for the homework, the quiz or the exam on which the infraction occurred***. Additional actions, including assigning the student a failing grade in the class or referring the case for disciplinary action, may be taken at the discretion of the instructors.

If you have any questions about labs or homework assignments, you may post them on Piazza. If you have a question concerning the specific code you wrote, ***please do not post your code publicly on Piazza***. Instead, make a private Piazza post, join office hours for help, or email the course instructors.

The following are considered violations of this course's academic integrity policy:

- Discussing code with others, showing someone else your code, or looking at someone else's code
- Having anyone else produce or fix code for you
- Copying code you find on the internet
- Using generative AI (e.g., ChatGPT) to produce code for you

You are not permitted to be in possession of any assignments, source code files, quizzes, or exams from another student either from the current semester or from past semesters. Possession of or sharing such files constitutes an infraction of

the academic integrity policy of this course. Penalty for copying files includes getting a zero for that assignment. You may also get reported to the university administration.

Your final grade will be based on (subject to change):

Labs	18% of grade (3% each for 6 labs)
Homework	36% of grade (6% each for 6 homework assignments)
Quizzes	20% of grade (4% each for 5 quizzes)
Final Exam	26% of grade

Lectures

Lectures are structured around introducing new data structures or algorithms, following which we will solve a problem together using the introduced concepts. Lectures are typically divided into modules, and for each module, we will provide partially- complete notes that you may print out and complete during class. Lectures will focus on both the conceptual and the implementation view of each data structure.

*Videos of lectures are prepared by CMU using Panopto software. These videos are for use **by students enrolled in the course and are not to be copied or made available to others.** If you ask questions in class, they may be captured on video. If you do not feel comfortable having your questions recorded, please talk to the instructors about other arrangements.* Students may not record lectures or labs without explicit permission from the instructors. Exceptions may be granted in accordance with university guidelines for accessibility concerns, but even then, such recordings may not be shared publicly.

Research on learning shows that unexpected noises and movement automatically divert and capture people's attention, which means you may be affecting everyone's learning experience if your cell phone, tablet, laptop, etc., makes noise or is visually distracting during class. ***For this reason, we allow you to take notes on your laptop, but ask that you turn the sound off and sit in the last row so that you do not disrupt other students' learning. You are not allowed to use your smartphone no matter where you sit.***

Labs

This course has weekly labs designed to review important concepts covered in lectures and apply them to solve a specific problem. In every lab, there will be questions that you will need to think about, and sample code may be provided to you. The focus here is more on the implementation view of data structures and using them to solve problems. **Please use Java 11 for your lab assignments.** Further, please do not put your code into public repositories (GitHub, BitBucket, etc.)

Course Website

As with many other CMU courses, we will use Canvas to host our course website. It is accessible via <https://canvas.cmu.edu>.

Posted on the course website will be:

- email addresses for the course staff,
- the schedule for office hours,
- copies of the lecture materials,
- example code files,
- homework and lab specifications.

Homework

This course has weekly homework assignments. More specifically, there will be 6 homework assignments that task you with solving problems using different data structures and algorithms, giving you chances to think creatively. **To mimic the real-world, the requirements (specifications) for assignments may not be 100% clear.** Please ask questions on Piazza and during office hours to make sure you get more information as needed. **As with labs, please use Java 11 for your homework, and do not put your code into public repositories or your own blogs.**

Unlike for labs, ***your highest score on Autolab is NOT final***. We will review your code and, if necessary, there will be deductions from that score. For every assignment, a standard rubric (outlining common issues and their associated deductions) will be used by the TAs when they grade. If you get 0 on Autolab, your score for the assignment will be 0. Since we need to manually grade your homework submissions, ***we will not accept late homework submissions***, except under exceptional cases (see the course policy on emergencies below). We strongly encourage you to start early so that you may have enough chances to get help, should you need it.

For both labs and homework assignments, you may use external resources (e.g., books, internet sites, including AI systems) as **references**, but you may not copy files, code, or text (or substantial parts thereof) from such resources. Also, **please do not upload course materials, assignment specs, or assignment solutions to websites (or systems) that will create derivative works. Specifically, this includes AI systems such as ChatGPT, GitHub Copilot, etc.**

Quizzes and Exam

There will be NO make-up quizzes and exams except for family, health, or personal emergencies. Please carefully read the policy on Family/Health/Personal Emergencies below.

- **Quizzes:** This course has weekly quizzes. The quizzes will be held during the last 15 to 20 minutes of a lecture period. The questions are meant to test not only your understanding of concepts but also your problem-solving skills. The schedule for quizzes has already been posted, and ***if you miss a quiz (outside of emergencies), you will receive a 0% for that quiz.***

- **Final exam:** The final exam will be a written exam covering all the materials in lectures, labs, quizzes, and homework assignments. More information will follow as we get closer to the exam. **The final exam is an 80-minute exam scheduled for May 5th, 2025.** Please plan your post-semester travel accordingly.
- Note: If your grade option is Pass/No Pass, you may be asked not to take the final exam if you are on track to receive a passing grade for the course.

Policy on Family/Health/Personal Emergencies

We understand that you may have emergencies occur that disrupt your ability to focus on learning. If you have a family/personal emergency or if you are too sick to engage with the course, please let us know.

We may need to confirm the emergency (e.g., with an email from your academic advisor for family/personal emergencies, or a note from your healthcare provider for illness); once it is confirmed, we're happy to arrange extensions for assignments and tests until you're ready to rejoin the course.

Note that we must be notified about emergencies before work is missed to provide extensions. Further, **job interviews are not considered emergencies. Please plan your interviews accordingly.**

Copyright Violations

You can take copies of course materials **only for personal use.**

- This applies especially to the lecture notes.
- This also applies to the course examples, quizzes, and the exam, and to all the homework assignments and lab write ups.
- Please do not make any course materials available to the public (especially on the Internet such as discussion forums or your blogs, etc.).
- Please do not put course materials into public repositories (e.g., GitHub).

No student may record or tape any classroom activity without the expressed written consent of the instructors.

Tentative Schedule

Week	Date	Modules	Quiz/Lab	Assignment
Week 1	03/11/2025	Module 1: Big Picture Module 2: Java Collections Module 3: Arrays and Linear Search		
	03/13/2025	Module 3: Arrays and Linear Search (cont'd) Module 4: ArrayList and Binary Search	Lab 1: ArrayList time comparison and String Manipulation	Homework 1 (due by 03/20)
Week 2	03/18/2025	Module 5: LinkedList	Sample Quiz	
	03/20/2025	Module 6: Stack Module 7: Queue	Lab 2: LinkedList Operation	Homework 2 (due by 03/27)
Week 3	03/25/2025	Module 8: Simple Sorting	Quiz 1	
	03/27/2025	Module 9: Comparable/Comparator Module 10: Recursion	Lab 3: Simple Sorting, Comparable/Comparator	Homework 3 (due by 04/08)
Week 4	04/01/2025	Module 10: Recursion (cont'd) Module 11: Hashing	Quiz 2	
	04/03/2025	No class!	Spring Carnival!	Enjoy!
Week 5	04/08/2025	Module 11: Hashing (cont'd) Module 12: HashTable	Lab 4: Recursion	Homework 4 (due by 04/15)
	04/10/2025	Module 13: HashSet, HashMap Module 14: Advanced Sorting	Quiz 3	
Week 6	04/15/2025	Module 14: Advanced Sorting (cont'd)	Lab 5: Advanced Sorting	Homework 5 (due by 04/22)
	04/17/2025	Module 15: Binary Trees	Quiz 4	
Week 7	04/22/2025	Module 15: Binary Trees (cont'd) Module 16: Huffman Coding Module 17: TreeSet, TreeMap	Lab 6: Binary Trees	Homework 6 (due by 04/29)
	04/24/2025	Module 18: Heaps, HeapSort	Quiz 5	
Week 8	05/05/2025	Final exam		

Accommodations for Students with Disabilities

If you have a disability and are registered with the Office of Disability Resources, we encourage you to use their online system to notify us of your accommodations and discuss your needs with us as early in the semester as possible. We will absolutely work with you to ensure that accommodations are provided as appropriate. If you suspect that you may have a disability and would benefit from accommodations but are not yet registered with the Office of Disability Resources, we encourage you to contact them at access@andrew.cmu.edu.

Statement of Support for Students Health & Well-being

Your health and well-being matters. Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress.

All of us benefit from support during times of struggle. You are not alone. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is often helpful. If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. Counseling and Psychological Services (CaPS) is here to help: call 412-268-2922 and visit their website at <http://www.cmu.edu/counseling/>. Consider reaching out to a friend, faculty, or family member you trust for help getting connected to the support that can help.

If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night:

CaPS: [412-268-2922](tel:412-268-2922)

Re:solve Crisis Network: [888-796-8226](tel:888-796-8226)

If the situation is life threatening, call the police:

On campus, CMU Police: [412-268-2323](tel:412-268-2323)

Off campus: 911